

Agri-Sync — Project Overview

What Is Agri-Sync?

Agri-Sync is a private, full-stack farm management system designed to digitise and centralise the tracking of field operations across multiple crop plots. It replaces manual record-keeping with structured data entry, automated calculations, and detailed analytics — giving farm managers a clear, real-time view of operations and costs at all times.

The system is self-hosted on the farm's private server, keeping all agricultural data internal and secure.

Purpose

The core purpose of Agri-Sync is to bridge the gap between what happens in the field and what management sees in the office.

Field technicians record operations directly on-site — including irrigation, fertilization, phytosanitary treatments, and harvesting — using a mobile-optimised app that works even without an internet connection. Those records are automatically synchronised to the central server when connectivity is available.

Managers then access a web dashboard where all recorded data is aggregated into structured reports, charts, and cost summaries — enabling informed decisions about resource usage, treatment schedules, and production costs per plot.

Key Characteristics

- Multi-crop:** The system supports multiple plots with different crop types and varieties, each tracked independently.
- Offline-first mobile access:** The field technician app functions fully without internet connectivity. Operations entered offline are queued locally and synced to the server automatically when a connection is restored.
- Automated calculations:** NPK (Nitrogen, Phosphorus, Potassium) nutrient units and production costs are calculated automatically based on configured fertilizer compositions, pesticide prices, water rates, and labor rates — no manual arithmetic required.
- Bilingual interface:** Both the mobile app and the admin dashboard support French and English. Users can switch languages at any time, and their preference is remembered.
- Role-based access:** The system enforces a hierarchical role model. Technicians can only enter data. Managers consult reports and view configuration in read-only mode. Administrators inherit all Technician and Manager permissions and additionally manage user accounts and system configuration.

Users

Role	Interface	Responsibilities
Field Technician	Mobile PWA	Records irrigation, fertilization, phytosanitary treatments, and harvesting operations per plot
Manager	Web dashboard	Consults reports, charts, and cost analytics per plot and per season; views configuration in read-only mode
Administrator	Web dashboard + Mobile PWA	Inherits all Technician and Manager permissions; additionally manages user accounts, plot definitions, and system configuration (fertilizers, pesticides, pricing)

Deployment

Agri-Sync is deployed entirely on the farm's **private server**. There is no reliance on third-party cloud services. The mobile app is distributed as a **Progressive Web App (PWA)** — field technicians access it directly from the browser on any device and can install it to their home screen without an app store.

System at a Glance

